Research Interests

Nonlinear Partial Differential Equations

Machine Learning, Deep Learning

Professional Experience

Research Scientist, Vianai Systems Inc., Palo Alto, CA, 2019-Present

Visiting Scholar, Department of Mathematics, University of Hartford, CT, 2020- Present.

Associate Professor, Department of Mathematics, University of Hartford, CT, 2017- 2020.

Assistant Professor, Department of Mathematics, University of Hartford, CT, 2011- 2017.

Teaching Assistant, Department of Mathematics, University of Kansas, KS, 2006-2011.

Research Assistant, Department, of Mathematics, University of Kansas, KS, 2005-2006.

Teaching Assistant, Department of Mathematics, Bogazici University, Turkey, 2004-2005.

Education

Ph.D. Mathematics, University of Kansas, 2005–2011.

M.S. Program Mathematics, Bogazici University, Turkey, 2004–2005.

B.S. Mathematics, METU, Turkey, 1999–2004.

B.S. Industrial Engineering, METU, Turkey, 1999–2004.

Profesional Certificates

Data Science Certificate Program, UC-Berkeley Extension, 2018-2019 (obtained during sabbatical leave).

Publications

1. RJ Decker, A Demirkaya, TJ Alexander, GA Tsolias, PG Kevrekidis, Fractional Solitons: A Homotopic Continuation from the Biharmonic to the Harmonic Model, Nonlinear Science, NLS-D-24-00109 (2025).

2. GA Tsolias, Robert J Decker, Aslihan Demirkaya, TJ Alexander, Ross Parker, PG Kevrekidis, Kink–antikink interaction forces and bound states in a nonlinear Schrödinger model with quadratic and quartic dispersion, Commun Nonlinear Sci Numer Simulat 125 (2023) 107362.

3. Tristram J Alexander, GA Tsolias, A Demirkaya, Robert J Decker, C Martijn de Sterke, PG Kevrekidis, Dark solitons under higher-order dispersion,Optics Letters, Vol.47, No. 5, 1174–1177 (2022).

4. G A Tsolias, R.J. Decker, A. Demirkaya, T. J. Alexander, P. G. Kevrekidis, Kink–antikink interaction forces and bound states in a ϕ^4 model with quadratic and quartic dispersion, Journal of Physics A: Mathematical and Theoretical, Vol. 54 (2021), No. 22.

5. I. C. Christov, R. Decker, A. Demirkaya, V.A. Gani, P.G. Kevrekidis, A. Saxena, Kink-Antikink, Collisions and Multi-Bounce Resonance Windows in Higher-Order Field Theories, arXiv:2005.00154 (2020).
6. R. Decker, A. Demirkaya, P.G. Kevrekidis, D. Iglesias, Y. Shavit, Kink Dynamics in a nonlinear Beam Model, Commun Nonlinear Sci Numer Simulat 97 (2021) 105747.
7. R. J. Decker, A. Demirkaya, N. S. Manton, and P.G. Kevrekidis, Kink-antikink interaction forces and bound states in a biharmonic ϕ^4- model, Journal of Physics A: Mathematical and Theoretical, Vol. 53 (2020), No. 37.
8. I. C. Christov, R. J. Decker, A. Demirkaya, V. A. Gani, P. G. Kevrekidis, Avinash Khare, Avadh Saxena, Kink-kink and kink-antikink interactions with long-range tails, Phys. Rev. Lett. 122, 1716019 (2019).
9. I. C. Christov, R. Decker, A. Demirkaya, P. G. Kevrekidis, V. A. Gani, R. V. Radomskiy, Long Range Interactions, Phys. Rev. D 99, 016010, (2019).
10. A. Demirkaya, R. Decker, P.G. Kevrekidis, I. Christov, A. Saxena, Kink Dynamics in a Parametric ϕ^6 Model, Journal of High Energy Physics (JHEP), DOI: 10.1007/JHEP12(2017)071.
11. A. Demirkaya, M. Stanislavova, Numerical results on existence and stability of standing and traveling waves for the fourth order beam equation, Journal Discrete and Continuous Dynamical System-B, (2017), 10.3934/dcdsb.2018097.
12. Hala Al-Khalil, C. Brennan, R. Decker, A. Demirkaya, J. Nagode, Numerical Existence and Stability of Steady State Solutions to the Distributed Spruce Budworm Model, Involve: A Journal of Mathematics, Vol. 10 (2017), No. 5, 857–879
13. A. Demirkaya, S. Hakkaev, On the spectral stability of periodic waves of the coupled Schrödinger equations, Physics Letters A, Vol 379 (2015), Issues 45-46, 2908-2914
14. A. Demirkaya, P.G. Kevrekidis, M. Stanislavova, A. Stefanov, Spectral stability analysis for standing waves of a perturbed Klein-Gordon equation, Dynamical Systems, Differential Equations and Applications AIMS Proceedings, (2015), 359-368
15. A. Demirkaya, S. Hakkaev, M. Stanislavova, A. Stefanov, On the spectral stability of periodic waves of the Klein-Gordon equation, Differential and Integral Equations, Vol 28 (2015), 431-454
16. A. Demirkaya, T. Kapitula, P.G. Kevrekidis, M. Stanislavova, A. Stefanov, On the spectral stability of kinks in some PT-symmetric variants of the classical Klein-Gordon Field Theories, Studies in Applied Mathematics, Vol. 133, Issue 3, pp.298-317, 2014
17. M. Aron, P. Bowers, N. Byer, R. Decker, A. Demirkaya, Jun Hwan Ryu, Numerical Results using Spectral Methods on Existence and Stability of Steady State Solutions for the Reaction Diffusion and Klein Gordon Equations, Involve: A Journal of Mathematics, Vol. 7, No. 6, pp.723-742, 2014
18. A. Demirkaya, D. J. Frantzeskakis, P. G. Kevrekidis, A. Saxena, A. Stefanov, Effects of PT -symmetry in Nonlinear Klein-Gordon Field Theories and Their Solitary Waves, Phys. Rev. E 88, 023203, 2013
19. A. Demirkaya, M. Stanislavova, Conditional stability theorem for the one dimensional Klein-Gordon equation, J. Math. Phys. 52, 112703, 2011.
20. A. Demirkaya, M. Stanislavova, Long Time Behavior for Radially Symmetric Solutions of Kuramoto-Sivashinsky Equation in Dimension 3, Dynamics of PDE, Vol. 7, No. 2, pp.161-173, 2010.
21. A. Demirkaya, The Existence of a Global Attractor for a Kuramoto-Sivashinsky type Equation in 2D, Discrete and Continuous Dynamical Systems-A, pp.198-207, 2009.

22. A. Demirkaya, L_2 estimates for the Solutions of Burger-Sivashinsky Equation in 1D and 2D, AIPS Conference Proceedings, Vol. 1184, pp.105-112, 2009.

Book Chapters

1. R.M. Ross, P.G. Kevrekidis, D. Campbell, A. Demirkaya, R. Decker, A Dynamical Perspective on the ϕ^4 Model: Past, Present and Future, edited by P. G. Kevrekidis and J. Cuevas-Maraver (Springer-Verlag, Heidelberg, 2019), Chap. 10.

Data Science Related Activities

1. May 20, 2022, WiDS Istanbul (@Sabanci University) Co-organizer and host to the panel "Data Science in Silicon Valley".
2. June 26th, 2021, WiDS Istanbul (@Uskudar University) Co-organizer and chair for the technical talks.
3. March 7th, 2021, WiDS Istanbul (@Sabanci University) Co-organizer and host to the panel "Data Science in USA".
4. 2020-present, WiDS Ambassador

Data Science Projects

1. Recognizing cities: Using Convolutional Deep Neural Networks, trained the machine to differentiate between the cities "Istanbul" and "Taipei" when a picture of those cities is given as an input.
2. Solving a partial differential equation (PDE) model using Python with Numpy and SciPy: Presented the numerical existence and the behavior of kink-antikink solutions of a well-known PDE.
3. Fraud detection on imbalanced data: Applied logistic regression, random forest, gbm models to a classification problem: credit card fraud detection. Tuned the models to achieve optimization.
4. Predicting house prices using lm and rpart models: Applied regression analysis using an OLSR linear (lm) model and a CART (rpart) model in predicting the house prices.
5. Bootstrapping and Permutation methods for significance testing: Using bootstrapping/permutation methods, constructed confidence intervals and null distribution for significance testing on the difference of mean incomes of males and females.
6. A/B testing using t-test, permutation test and logistic regression: Using A/B testing, studied if there is a significant difference between mean daily returns of Google stock and S&P 500.
7. Clustering Companies Based Stock Price Movements: Applied PCA, TSNE, K-means and hierarchical clustering, clustered the records so that the companies in the same industry are grouped together.

Invited Talks

May 14-15, 2025, University of Alabama-Birmingham, Third Joint Alabama-Florida Conference, Panelist.

Nov 2, 2024, University of Washington, Shaping Careers: Applying Math Expertise in Data Science/AI during Academia to Industry Transition

Aug 15, 2023, DRP Math Turkiye, Applying Math Expertise in AI during the Academia-to-Industry Transition

May–23-27, 2021, SIAM Conference on Applications of Dynamical Systems, Virtual Conference.

November 14, 2020, Panelist at KU Math in Industry Careers, Virtual Event.

September 13, 2020, AFLIM Webinar, Virtual Event.

March 2, 2020, WiDS Stanford, Panelist at the Ethics Panel, Stanford University, CA.

May–19-23, 2019, SIAM Conference on Applications of Dynamical Systems, Snowbird, Utah.

March 25, 2019, St Mary's College of California, Moraga, CA.

June–11-14, 2018, SIAM Conference on Nonlinear Waves and Coherent Structures, Anaheim, CA.

March 17–18, 2018, AMS Sectional Meeting, Columbus, Ohio.

June 18–23, 2017, CMO-BIRS, Geometrical Methods, non Self-Adjoint Spectral Problems, and Stability of Periodic Structures, Oaxaca, Mexico.

March 4, 2017, WINRS Conference, Brown University, Providence, MA.

September 30 -October 2, 2016, 2nd Annual Meeting of the SIAM Central States Section, University of Arkansas, Arkansas.

September 15-21, 2013, Oberwolfach Workshop, Lattice Differential Equations, Oberwolfach, Germany

May 22-26, 2011, SIAM Conference on Applications of Dynamical Systems, Snowbird, Utah.

March 19-20, 2011, 67th Midwest Partial Differential Equations, University of Illinois Urbana-Champaign.

March 4, 2011, University of Hartford, USA.

May 2010, 8th AIMS International conference, Dresden University of Technology, Dresden, Germany.

April 16-18, 2010, Conference on Harmonic Analysis and PDEs, University of Nebraska-Lincoln.

November 21-22, 2009, The First Oklahoma PDE Workshop, Oklahoma State University.

June 7-12, 2009, 35th International Conference Applications of Mathematics in Engineering and Economics, Sozopol, Bulgaria.

June 2, 2009, Sofia University, Sofia, Bulgaria.

March 23-26, 2009, 6th IMACS International Conference on Nonlinear Evolution Equations and Wave Phenomena at Athens, Georgia, USA.

May 18-21, 2008, 7th AIMS International Conference on Dynamical Systems, Differential Equations and Applications University of Texas at Arlington.

November 2-3, 2007, Seventh Prairie Analysis Seminar, Kansas State University.

Poster Presentations

April 21-23, 2017, KUMUNU Conference in PDE, Dynamical Systems and Applications, University of Nebraska-Lincoln.

April 23-24, 2016, KUMU Conference on PDE, Dynamical Systems, and Applications, University of Missouri-Columbia

March 30-April 1, 2012, AMS Spring Central Section Meeting, University of Kansas.

Jan 9, 2011, Joint Meetings, New Orleans

Undergraduate Research

Co-supervised Projects

1. Spring 2018, Yonathan Shavit, Jeffey Severino, Javier Melecio, Digno Iglesias, *Kink Dynamics in a nonlinear Beam Model.*
2. Spring 2017, Robert Galvez, Matthew Bernocco, *Soliton interaction for a nonlinear Neuron Equation.*
3. Spring 2017, Iliana Albion-Poles, Mitchell Sugar, Yonatan Shavit, *Soliton escape velocity after scattering for a nonlinear Beam Equation.*
4. Spring 2016, Salem Moges and Reid Bassette ,*Soliton escape velocity after scattering for a nonlinear Klein-Gordon partial differential equation.*
5. Fall/Spring 2016, Damaris Zachos, Gianmarco Molino, *Comparison of KDV and RLW PDE models of shallow-water waves.*
6. Spring 2015,Steven Kingston, Jarrett Lagler, and Gianmarco Molino, *Acoustic Waves in Neurons.*
7. Spring 2014, Hala Al-Khalil and Jamie Nagode, *Numerical Existence and Stability of Traveling Wave Solutions to the Distributed Spruce Budworm Model.*
8. Spring 2014, Ethan Bourdeau, John Cunsole, Virginia Demske, and Scott Rubin, *Numerical Existence, Stability and Simulation of Solitons to the Sine-Gordon Equation.*
9. Spring 2013, Gino Cordone and Jeffrey Knecht, *Existence of Period-Doubling and Chaos of nonlinear forced Klein-Gordon equation.*
10. Spring 2013, Catherine Brennan and James Pellissier, *Numerical Existence and Stability of Traveling Wave Solutions to the Distributed Spruce Budworm Model.*
11. Spring 2013, Jessica Andersen and Cole Murphy, *The forced Van der Pol Equation.*
12. Spring 2013, David Arena, Karen Brzostowski, Adam Paul, and Christopher Vincent, *Differential Equation Manipulation using a Matlab Graphical User Interface.*
13. Spring 2012, Nicole Byer and Jun Hwan Ryu, *The numerical existence and stability of the steady state solutions of the Reaction-Diffusion equation.*
14. Spring 2012, Miles Aron, Peter Bowers, *The numerical existence and stability of the steady state solutions of the Klein-Gordon equations.*

Undergraduate research conference talks

April 28, 2018, The Spuyten Duyvil Undergraduate Mathematics Conference, Southern Connecticut State University, CT.

April 8, 2017, Hudson River Undergrad Math Conference, Westfield State University, Westfield, MA

April 2nd, 2016, The Hudson River Undergraduate Mathematics Conference (HRUMC), St Michael's College, Colchester, VT.

April 18th, 2015, 10th Annual Spuyten Duyvil Undergraduate Mathematics Conference, Manhattan College, Riverdale, NY.

April 26th, 2014, The Joint Hudson River and Spuyten Duyvil Undergraduate Mathematics Conference, Marist College in Poughkeepsie, NY

April 6th, 2013, 20th Hudson River Undergraduate Mathematics Conference (HRUMC), Williams College in Williamstown, MA.

April 21st, 2012, 19th Hudson River Undergraduate Mathematics Conference, Western New England University, Springfield, MA.

April 14, 2012, 7th Annual Spuyten Duyvil Undergraduate Mathematics Conference, Ramapo College of New Jersey.

Student Research Funds

March 2018, Dean's Research Funds, granted from the Office of the Dean of Arts and Sciences. ($200)

March 2017, Dean's Research Funds, granted from the Office of the Dean of Arts and Sciences. ($300)

March 2016, Dean's Research Funds, granted from the Office of the Dean of Arts and Sciences. ($280)

March 2015, Dean's Research Funds, granted from the Office of the Dean of Arts and Sciences. ($350)

March 2014, Dean's Research Funds, granted from the Office of the Dean of Arts and Sciences. ($400)

March 2013, Dean's Research Funds, granted from the Office of the Dean of Arts and Sciences. ($1700)

March 2012, Dean's Research Funds, granted from the Office of the Dean of Arts and Sciences. ($750)

Membership in undergraduate examination committees

April 2014, Catherine Brennan, member of honor thesis committee

Teaching Experience

Assistant Professor at the University of Hartford

DIA 100 Freshman Dialog, Fall 2015, Fall 2013

M114 Everyday Statistics, Spring 2018, Fall 2017, Spring 2017

M140 PreCalculus, Fall 2011

M144 Calculus I, Spring 2012

M145 Calculus II, Spring 2013

M220 Linear Algebra & Matrix Theory, Spring 2016, Spring 2015

M242 Ordinary Differential Equations, Fall 2017, Fall 2016, Summer 2016, Fall 2015, Spring 2014, Fall 2013, Spring 2013, Fall 2012, Spring 2012

M340 Introductory Analysis, Fall 2016, Fall 2015, Fall 2013

M344 Advanced Engineering Mathematics, Spring 2018, Fall 2017, Spring 2017, Spring 2016, Fall 2016, Fall 2015, Spring 2015, Spring 2014, Fall 2013, Spring 2013, Fall 2012, Fall 2011

M480 Independent Study Mathematics, Spring 2018, Fall 2017, Spring 2017, Spring 2016, Fall 2015, Spring 2015, Spring 2014, Fall 2013, Spring 2013, Fall 2012, Spring 2012

Teaching Assistant at the University of Kansas

Math 121, Calculus II for Engineering Students, Spring 2011

Math 220, Applied ODEs, Fall 2010

Instructor at the University of Kansas

Math 290, Elementary Linear Algebra, Spring 2011

Math 121, Calculus I for Engineering Students, Spring 2010, Fall 2009, Spring 2008, Fall 2007, Spring 2007

Math 122, Calculus II for Engineering Students, Spring 2009, Fall 2008

Math 115, Calculus for Business School Students, Fall 2006, Summer 2006, Spring 2006

Teaching Assistant at Bogazici University

Math 202, Differential Equations, Spring 2005

Math 132, Calculus II for Mathematics Students, Fall 2004

Math 231, Calculus III for Mathematics Students, Fall 2004

Honors and Awards

2017, Dean's Research Funds, granted from the Office of the Dean of Arts and Sciences. ($592)

2017, Travel Award, KUMUNU Conference on PDE, Dynamical Systems, and Applications ($692)

2016, Travel Award, KUMU Conference on PDE, Dynamical Systems, and Applications ($400)

2013, OWLG grant, Oberwolfach Workshop, Lattice Differential Equations, Oberwolfach ($500)

2013, Travel Award, IMA Special Workshop, Joint US-Japan Conference for Young Researchers on Interactions among Localized Patterns in Dissipative Systems ($700)

2011, Joint Meetings New Orleans, AWM travel award ($800)

2010-2011, KU Women of Distinction, University of Kansas

2010, Summer Research Fellowship, Graduate Studies, University of Kansas ($5000)

2010, Phi Beta Delta Honor Society, KU Chapter, Outstanding International Student Award

2010 AIMS International Conference, NSF travel award ($750)

2009 June 7-12, 35th AMEE, Sozopol, Bulgaria, Best Presentation of the paper entitled L_2 estimates for the Solutions of Burger-Sivashinsky Equation in 1D and 2D

2009 Summer, Black-Landis Scholarship, University of Kansas

2008 Nonlinear Waves '08 workshop, Rome, Italy, NSF travel award ($500)

2008 Summer School, $1000 (for travel expenses), Clay Mathematics Institute

2008 Summer, The Paul E. Conrad Scholarship, University of Kansas

2008 Summer, Black-Mitchell Scholarship, University of Kansas

2008 SIAM Conference on Dynamical Systems, NSF travel award ($400)

2007 Summer, Black-Mitchell Scholarship, University of Kansas

2007 Spring, Joan Kirkham Travel Scholarship, University of Kansas

2005 Fall, Research Assistantship, University of Kansas

2002-2004, Undergraduate Math Scholarship of the Scientific and Technological Research Council of Turkey

1999-2004, Dean's Honors list, METU, Ankara, Turkey

1999, Top 0.1% in Turkey, University Entrance Examination

1999, Bronze Medal, National Mathematics Olympiads, Turkey

1998, Ranked 1st in Marmara Region, National Mathematics Olympiads, Turkey

1996, Silver Medal, National Mathematics Olympiads, Turkey

Training and seminars

2016 Feb-May, Nonlinear Waves Working Group Seminar, UMass-Amherst

2013 June 3-7, IMA Special Workshop, Joint US-Japan Conference for Young Researchers on Interactions among Localized Patterns in Dissipative Systems

2008 July 21- 24, SIAM Conference on Nonlinear Waves and Coherent Structures, University of Rome, Italy

2008 July 18- 19, Multidimensional Localized Structures, University of Rome, Italy

2008 June 23-July 18, Clay Mathematics Institute Summer School on Evolution Equations, ETH Swiss Federal Institute of Technology in Zurich, Switzerland

2007 January 24-26, MSRI Dynamical Systems Workshop, San Francisco, California

2006 October 13-14, Sixth Prairie Analysis Seminar, University of Kansas, Lawrence, Kansas

2005 October 14-15, Fifth Prairie Analysis Seminar, Kansas State University, Manhattan, Kansas

Scholarly Service

Referee for Mediterranean Journal of Mathematics

Referee for European Journal of Applied Mathematics

Referee for Studies In Applied Mathematics

Referee for Advances in Difference Equations

Referee for Communications on Pure & Applied Analysis

Departmental and Campus Service

University of Hartford

2018, April-Member of Greenberg Junior Faculty Grant Committee

2018, April-Math and Statistics Awareness Month, Faculty Advisor

2018, April-SCUDEM Faculty Coach

2018, Member of the Hiring Committee of Computer Science Department

2017-2019 Advisory Board member of Students Achieving Success (SAS).

2017-2018, Member of the Hiring Committee of Math Department

2017-Fall Preview Panel, Faculty Panelist

2016-2018 Member of the Academic Standing Committee

2015-2018 Member of the Curriculum Committee

2016-2017 Member of the Senior Regents Award Committee

2013-2018 Faculty Advisor (with Mark Turpin) for Math Club

2015-2016, Advisor for 13 CS students

2014-2016, Advisor for 2 Math students

2013-2014, Advisor for 7 Math/Math Ed students

2012, Faculty Marshal for the College of Arts and Sciences

University of Kansas

2009-2010, President of KU Student Chapter of Association for Women in Mathematics (AWM)

2009-2010, Vice President, International Student of the KU Chapter of the Phi Beta Delta

2007-2010, April, Math Awareness Month presenter

2007-2010, International Math competition Math Kangaroo assistant

2007-2010, April, Mathematics Math Awareness Math competition assistant

2008-2009, Student Representative on Graduate Committee

2007-2008, President of Mathematics Graduate Student Organization (GSO), University of Kansas

2006-2007, President of Turkish Student Organization

2006-2007, Treasurer of Mathematics Graduate Student Organization (GSO), University of Kansas

2006-2007, Treasurer of KU Student Chapter of Association for Women in Mathematics (AWM)

Community Outreach

Students Achieving Success (SAS) Advisory Board member, April 2017-present.

MATHCOUNT volunteer, 2017 Hartford Chapter

Feb 4th, Scorer for Regional Competition
March 11th, Scorer for State Competition

Memberships

2006-present, Association for Women in Mathematics

2006-present, American Mathematical Society

2005-present, Turkish American Association

2003-present, Society for Industrial and Applied Mathematics